

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
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Originating Company	Epic Atlantic Limited
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INSTRUMENT MTOs

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Revision History

R01	28.06.22	Issued for Review	P.Roy	P.Olukah			
Rev No	Date	Reason for Issue / Change	Prepared	Reviewed	TCE Approval		

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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Table of Contents

Transmitter MTO	4
SDV BDV MTO	5
Local Guages	6
FGS MTO	7

Sr. No.	Tag No.	Process Description	Type of Instrument	Line Rating	Line Pressure Range	Type of Sensor	Enclosure Material	Certifications	Output	Wetted Part MOC	Mounting	Accessories	Make	Model
1	BBOU-DPT-1001	Differential Pressure Measurement at the inlet across Venturi Meter	DP Transmitter	#900	0- 5000 mmWC	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
2	BBOU-PIT-1001	Battery Limit Pressure at the Inlet Manifold of BBOU	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
3	BBOU-TIT-1001	Battery Limit Temperature at the Inlet Manifold of BBOU	Temperature Transmitter	#900	0-100 degC	RTD PT 100 Input	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	-	Direct	M20 Gland,External SPD	Emerson	644
4	BBOU-PIT-1002	Flowline Pressure for interlock-BBOU Gathering Manifold	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
5	BBOU-PIT-1003	Flowline Pressure for interlock-BBOU Gathering Manifold	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
6	BBOU-PIT-1004	Flowline Pressure for interlock-BBOU Gathering Manifold	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
7	BBOU-PIT-1005	Instrument Gas Manifold Pressure Measurment	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
8	BBOU-PIT-1013	Flowline Pressure for interlock-BBOU-A-200 Launcher	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
9	BBOU-PIT-1014	Flowline Pressure for interlock-BBOU-A-200 Launcher	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
10	BBOU-PIT-1015	Flowline Pressure for interlock-BBOU-A-200 Launcher	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
11	BBOU-PIT-1016	Pressure measurement of closed Drain Vessel BBOU-CDT-001	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
12	BBOU-LIT-1001	Level for interlock-Closed Drain Vessel BBOU-CDT-001	Guided Wave Radar Transmitter	#150	0-2400mm	GWR Radar	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	M20 Gland,External SPD	Emerson	5300
13	BBOU-TIT-1003	Temperature on the Closed Drain Vessel BBOU-CDT-001	Temperature Transmitter	#900	0-100 degC	RTD PT 100 Input	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	-	Direct	M20 Gland,External SPD	Emerson	644
14	BBOU-LIT-1002	Level at BBOU-ODT-001	Guided Wave Radar Transmitter	#150	0-2400mm	GWR Radar	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	M20 Gland,External SPD	Emerson	5300
15	ELEP-DPT-1001	Differential Pressure Measurement at the inlet across Venturi Meter	DP Transmitter	#900	0- 5000 mmWC	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
16	ELEP-PIT-1001	Battery Limit Pressure at the Inlet Manifold of ELEP	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
17	ELEP-TIT-1001	Battery Limit Temperature at the Inlet Manifold of ELEP	Temperature Transmitter	#900	0-100 degC	RTD PT 100 Input	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	-	Direct	M20 Gland,External SPD	Emerson	644
18	ELEP-PIT-1002	Flowline Pressure for interlock-ELEP Gathering Manifold	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
19	ELEP-PIT-1003	Flowline Pressure for interlock-ELEP Gathering Manifold	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
20	ELEP-PIT-1004	Flowline Pressure for interlock-ELEP Gathering Manifold	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
21	ELEP-PIT-1005	Instrument Gas Manifold Pressure Measurment	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
22	ELEP-PIT-1010	Battery Limit Pressure at the Inlet of Pig Receiver ELEP-A-200	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
23	ELEP-TIT-1003	Battery Limit Temperature-Pig Receiver ELEP-A-200	Temperature Transmitter	#900	0-100 degC	RTD PT 100 Input	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	-	Direct	M20 Gland,External SPD	Emerson	644
24	ELEP-PIT-1018	Flowline Pressure for interlock-ELEP-A-300 Launcher	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
25	ELEP-PIT-1019	Flowline Pressure for interlock-ELEP-A-300 Launcher	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
26	ELEP-PIT-1020	Flowline Pressure for interlock-ELEP-A-300 Launcher	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
27	ELEP-PIT-1021	Pressure measurement of closed Drain Vessel ELEP-CDT-001	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
28	ELEP-LIT-1001	Level for interlock-Closed Drain Vessel ELEP-CDT-001	Guided Wave Radar Transmitter	#150	0-2400mm	GWR Radar	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	M20 Gland,External SPD	Emerson	5300
29	ELEP-TIT-1004	Temperature on the Closed Drain Vessel ELEP-CDT-001	Temperature Transmitter	#900	0-100 degC	RTD PT 100 Input	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	-	Direct	M20 Gland,External SPD	Emerson	644
30	ELEP-LIT-1002	Level at ELEP-ODT-001	Guided Wave Radar Transmitter	#150	0-2400mm	GWR Radar	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	M20 Gland,External SPD	Emerson	5300
31	AKAB-PIT-1001	Battery Limit Pressure at the Inlet of Pig Receiver AKAB-A-100	Pressure Transmitter	#900	0-100 barg	Piezoelectric	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	SS316	Direct	Monoflange Manifold, M20 Gland, External SPD	Emerson	3051T
32	AKAB-TIT-1001	Battery Limit Temperature-Pig Receiver AKAB-A-100	Temperature Transmitter	#900	0-100 degC	RTD PT 100 Input	DCA with Low copper	IIA/IIIB, T4 (Exia,Exd) IP-65, SIL-2	4-20 mA/HART	-	Direct	M20 Gland,External SPD	Emerson	644

Sr. No.	Tag No.	Application	Type of Valve	Line Size	Rating	Actuator Sizing Pressure (Shut off Pressure)	Body MOC	Ball and Trim	Shut off Class	Safety Margin (SST/VsCT)	Area Classification	Solenoid	Limit Switch	Fail Position	Make
1	BBOU-SDV-1001	Inlet Shutdown Valve-BBOU Gathering Manifold	Shutdown Valve- Gas Over Liquid	24"	#900 RF	110 Barg	A 105	SS316 Stellite	TSO	2	IIA,IIB , T3, IP 65	2 Nos (Exd,24vDC)	Micro Switch	FC	Cameron
2	BBOU-SDV-1002	Blowdown Valve to Vent	Blowdown Valve- Gas Over Liquid	4"	#900 RF	110 Barg	A 105	SS316 Stellite	TSO	2	IIA,IIB , T3, IP 65	2 Nos (Exd,24vDC)	Micro Switch	FO	Cameron
3	BBOU-SDV-1003	Outlet Shutdown Valve- Pig Launcher BBOU-A-200	Shutdown Valve- Gas Over Liquid	24"	#900 RF	110 Barg	A 105	SS316 Stellite	TSO	2	IIA,IIB , T3, IP 65	2 Nos (Exd,24vDC)	Micro Switch	FC	Cameron
4	ELEP-SDV-1001	Inlet Shutdown Valve-ELEP Gathering Manifold	Shutdown Valve- Gas Over Liquid	36"	#900 RF	110 Barg	A 105	SS316 Stellite	TSO	2	IIA,IIB , T3, IP 65	2 Nos (Exd,24vDC)	Micro Switch	FC	Cameron
5	ELEP-SDV-1002	Blowdown Valve to Vent	Blowdown Valve- Gas Over Liquid	4"	#900 RF	110 Barg	A 105	SS316 Stellite	TSO	2	IIA,IIB , T3, IP 65	2 Nos (Exd,24vDC)	Micro Switch	FO	Cameron
6	ELEP-SDV-1003	Inlet Shutdown Valve-OPUGBENE Pig Receiver ELEP-A-200	Shutdown Valve- Gas Over Liquid	24"	#900 RF	110 Barg	A 105	SS316 Stellite	TSO	2	IIA,IIB , T3, IP 65	2 Nos (Exd,24vDC)	Micro Switch	FC	Cameron
7	ELEP-SDV-1004	Outlet Shutdown Valve- Pig Launcher ELEP-A-300	Shutdown Valve- Gas Over Liquid	36"	#900 RF	110 Barg	A 105	SS316 Stellite	TSO	2	IIA,IIB , T3, IP 65	2 Nos (Exd,24vDC)	Micro Switch	FC	Cameron
8	AKAB-SDV-1003	Inlet Shutdown Valve-OPUGBENE Pig Receiver AKAB-A-100	Shutdown Valve- Gas Over Liquid	36"	#900 RF	110 Barg	A 105	SS316 Stellite	TSO	2	IIA,IIB , T3, IP 65	2 Nos (Exd,24vDC)	Micro Switch	FC	Cameron

Sr. No.	Tag No.	Application	Type of instrument	Line Size	Line Rating	Pipe Flow Capacity	Pressure Range	Instrument Range	MOC	Protection	Accessories
1	BBOU-FE/FT-1001	Check Meter at the inlet manifold	Venturi Meter	24"	900# RF	0-331 mmscfd	0-110 Barg	0-400 mmscfd	SS Wetted Parts	-	-
2	BBOU-PI-1001	Battery Limit Pressure at the Inlet Manifold of BBOU	Pressure Gauge	24"	900# RF	-	0-110 Barg	0-160 barg	SS Wetted Parts	IP-66	Solid Front
3	BBOU-TW/TE-1001	Battery Limit Temperature at the Inlet Manifold of BBOU	Temperature Gauge with Thermowell	24"	900# RF	-	0-110 Barg	0-100 degC	SS316 Thermowell	IP-66	Thermowell Length 300 mm
4	BBOU-TW/TE-1001	Battery Limit Temperature at the Inlet Manifold of BBOU	RTD with Thermowell	24"	900# RF	-	0-110 Barg	0-100 degC	SS316 Thermowell	IP-66, Exd IIA, IIB, T4	Thermowell Length 300 mm
5	BBOU-PI-1002	Flowline Pressure to BBOU-A-200 Launcher	Pressure Gauge	24"	900# RF	-	0-110 Barg	0-160 barg	SS Wetted Parts	IP-66	Solid Front
6	BBOU-RO-1001	Restriction Orifice to Vent-BBOU Vent Line Depressurisation Orifice	Restriction Orifice	4"	900# RF	-	0-110 Barg	-	SS Wetted Parts	-	-
7	BBOU-PI-1010	Pressure on the Pig Launcher BBOU-A-200	Pressure Gauge	-	900# RF	-	0-110 Barg	0-160 barg	SS Wetted Parts	IP-66	Solid Front
8	BBOU-PI-1011	Pressure on the Pig Launcher BBOU-A-200	Pressure Gauge	-	900# RF	-	0-110 Barg	0-1 barg	SS Wetted Parts	IP-66	Solid Front, gauge saver
9	BBOU-PI-1012	Pressure at the outlet of the Pig Launcher BBOU-A-200	Pressure Gauge	24"	900# RF	-	0-110 Barg	0-160 barg	SS Wetted Parts	IP-66	Solid Front
10	BBOU-RO-1003	Restriction Orifice to Vent - BBOU-A-200	Restriction Orifice	4"	900# RF	-	0-110 Barg	-	SS Wetted Parts	-	-
11	BBOU-LG-1001	Level Indication at Closed Drain Vessel BBOU-CDT-001	Level Gauge	-	150#RF	-	-	0-2400 mm	SS316	-	Reflex Gauge
12	BBOU-TW/TE-1003	Temperature on the Closed Drain Vessel BBOU-CDT-001	RTD with Thermowell	-	150#RF	-	0-10 Barg	0-100 degC	SS316 Thermowell	IP-66, Exd IIA, IIB, T4	Thermowell Length 300 mm
13	ELEP-FE/FT-1001	Check Meter at the inlet manifold	Venturi Meter	36"	900# RF	0-340mmscfd	0-110 Barg	0-400 mmscfd	SS Wetted Parts	-	-
14	ELEP-PI-1001	Battery Limit Pressure at the Inlet Manifold of ELEP	Pressure Gauge	36"	900# RF	-	0-110 Barg	0-160 barg	SS Wetted Parts	IP-66	Solid Front
15	ELEP-TW/TE-1001	Battery Limit Temperature at the Inlet Manifold of ELEP	Temperature Gauge with Thermowell	36"	900# RF	-	0-110 Barg	0-100 degC	SS316 Thermowell	IP-66	Thermowell Length 300 mm
16	ELEP-TW/TE-1001	Battery Limit Temperature at the Inlet Manifold of ELEP	RTD with Thermowell	36"	900# RF	-	0-110 Barg	0-100 degC	SS316 Thermowell	IP-66, Exd IIA, IIB, T4	Thermowell Length 300 mm
17	ELEP-PI-1002	Flowline Pressure to ELEP-A-200 Launcher	Pressure Gauge	36"	900# RF	-	0-110 Barg	0-160 barg	SS Wetted Parts	IP-66	Solid Front
18	ELEP-RO-1001	Restriction Orifice to Vent-ELEP Vent Line Depressurisation Orifice	Restriction Orifice	4"	900# RF	-	0-110 Barg	-	SS Wetted Parts	-	-
19	ELEP-PI-1010	Battery Limit Pressure at the Inlet of Pig Receiver ELEP-A-200	Pressure Gauge	24"	900# RF	-	0-110 Barg	0-160 barg	SS Wetted Parts	IP-66	Solid Front
20	ELEP-TW/TE-1003	Battery Limit Temperature-Pig Receiver ELEP-A-200	Temperature Gauge with Thermowell	24"	900# RF	-	0-110 Barg	0-100 degC	SS316 Thermowell	IP-66	Thermowell Length 300 mm
21	ELEP-TW/TE-1003	Battery Limit Temperature-Pig Receiver ELEP-A-200	RTD with Thermowell	24"	900# RF	-	0-110 Barg	0-100 degC	SS316 Thermowell	IP-66, Exd IIA, IIB, T4	Thermowell Length 300 mm
22	ELEP-PI-1011	Pressure at the Pig Receiver ELEP-A-200	Pressure Gauge	-	900# RF	-	0-110 Barg	0-160 barg	SS Wetted Parts	IP-66	Solid Front
23	ELEP-PI-1012	Pressure on the Pig Receiver ELEP-A-200	Pressure Gauge	-	900# RF	-	0-110 Barg	0-160 barg	SS Wetted Parts	IP-66	Solid Front
24	ELEP-PI-1013	Pressure on the Pig Receiver ELEP-A-200	Pressure Gauge	-	900# RF	-	0-110 Barg	0-1 barg	SS Wetted Parts	IP-66	Solid Front, gauge saver
25	ELEP-RO-1003	Restriction Orifice to Vent - ELEP-A-200	Restriction Orifice	4"	900# RF	-	0-110 Barg	-	SS Wetted Parts	-	-
26	ELEP-PI-1014	Pressure on the Pig Launcher ELEP-A-300	Pressure Gauge	-	900# RF	-	0-110 Barg	0-160 barg	SS Wetted Parts	IP-66	Solid Front
27	ELEP-PI-1015	Pressure on the Pig Launcher ELEP-A-300	Pressure Gauge	-	900# RF	-	0-110 Barg	0-1 barg	SS Wetted Parts	IP-66	Solid Front, gauge saver
28	ELEP-PI-1016	Pressure at the outlet of the Pig Launcher ELEP-A-300	Pressure Gauge	36"	900# RF	-	0-110 Barg	0-160 barg	SS Wetted Parts	IP-66	Solid Front
29	ELEP-RO-1004	Restriction Orifice to Vent - ELEP-A-300	Restriction Orifice	4"	900# RF	-	0-110 Barg	-	SS Wetted Parts	-	-
30	ELEP-LG-1001	Level Indication at Closed Drain Vessel ELEP-CDT-001	Level Gauge	-	150#RF	-	-	0-2400 mm	SS316	-	Reflex Gauge
31	ELEP-TW/TE-1003	Temperature on the Closed Drain Vessel ELEP-CDT-001	RTD with Thermowell	-	150#RF	-	0-10 Barg	0-100 degC	SS316 Thermowell	IP-66, Exd IIA, IIB, T4	Thermowell Length 300 mm
32	AKAB-PI-1001	Battery Limit Pressure at the Inlet of Pig Receiver AKAB-A-100	Pressure Gauge	36"	900# RF	-	0-110 Barg	0-160 barg	SS Wetted Parts	IP-66	Solid Front
33	AKAB-TW/TE-1001	Battery Limit Temperature-Pig Receiver AKAB-A-100	Temperature Gauge with Thermowell	36"	900# RF	-	0-110 Barg	0-100 degC	SS316 Thermowell	IP-66	Thermowell Length 300 mm
34	AKAB-TW/TE-1001	Battery Limit Temperature-Pig Receiver AKAB-A-100	RTD with Thermowell	36"	900# RF	-	0-110 Barg	0-100 degC	SS316 Thermowell	IP-66, Exd IIA, IIB, T4	Thermowell Length 300 mm
35	AKAB-PI-1011	Pressure at the Pig Receiver AKAB-A-100	Pressure Gauge	-	900# RF	-	0-110 Barg	0-160 barg	SS Wetted Parts	IP-66	Solid Front
36	AKAB-PI-1012	Pressure on the Pig Receiver AKAB-A-100	Pressure Gauge	-	900# RF	-	0-110 Barg	0-160 barg	SS Wetted Parts	IP-66	Solid Front
37	AKAB-PI-1013	Pressure on the Pig Receiver AKAB-A-100	Pressure Gauge	-	900# RF	-	0-110 Barg	0-1 barg	SS Wetted Parts	IP-66	Solid Front, gauge saver
38	AKAB-RO-1003	Restriction Orifice to Vent - AKAB-A-100	Restriction Orifice	4"	900# RF	-	0-110 Barg	-	SS Wetted Parts	-	-

Sr. No.	Tag No.	Type of Instrument	Type of Detector
1	BBOU-AE/AT-1001	Point Type Gas Detector	IR Type HC Detector
2	BBOU-AE/AT-1002	Point Type Gas Detector	IR Type HC Detector
3	BBOU-AE/AT-1003T	Open Path Gas Detector (Transmitter)	IR Type Open Path HC Detector-Transmitter
4	BBOU-AE/AT-1003R	Open Path Gas Detector (Receiver)	IR Type Open Path HC Detector-Receiver
5	BBOU-AE/AT-1004T	Open Path Gas Detector (Transmitter)	IR Type Open Path HC Detector-Transmitter
6	BBOU-AE/AT-1004R	Open Path Gas Detector (Receiver)	IR Type Open Path HC Detector-Receiver
7	BBOU-MAC-1001A	Manual Alarm Call Point (Field ESD)	Manual Call Point
8	BBOU-MAC-1002A	Manual Alarm Call Point (Field ESD)	Manual Call Point
9	BBOU-MAC-1003A	Manual Alarm Call Point (Field ESD)	Manual Call Point
10	BBOU-MAC-1004A	Manual Alarm Call Point (Field ESD)	Manual Call Point
11	BBOU-HB-1004	Hooter and beacon combine	Hooter-Beacon
12	ELEP-AE/AT-1001	Point Type Gas Detector	IR Type HC Detector
13	ELEP-AE/AT-1002	Point Type Gas Detector	IR Type HC Detector
14	ELEP-AE/AT-1003T	Open Path Gas Detector (Transmitter)	IR Type Open Path HC Detector-Transmitter
15	ELEP-AE/AT-1003R	Open Path Gas Detector (Receiver)	IR Type Open Path HC Detector-Receiver
16	ELEP-AE/AT-1004T	Open Path Gas Detector (Transmitter)	IR Type Open Path HC Detector-Transmitter
17	ELEP-AE/AT-1004R	Open Path Gas Detector (Receiver)	IR Type Open Path HC Detector-Receiver
18	ELEP-MAC-1001A	Manual Alarm Call Point (Field ESD)	Manual Call Point
19	ELEP-MAC-1002A	Manual Alarm Call Point (Field ESD)	Manual Call Point
20	ELEP-MAC-1003A	Manual Alarm Call Point (Field ESD)	Manual Call Point
21	ELEP-MAC-1004A	Manual Alarm Call Point (Field ESD)	Manual Call Point
22	ELEP-HB-1004	Hooter and beacon combine	Hooter-Beacon
23	BBOU-AE/AT-1003	Point Type Gas Detector	IR Type HC Detector
24	BBOU-AE/AT-1004T	Open Path Gas Detector (Transmitter)	IR Type Open Path HC Detector-Transmitter
25	BBOU-AE/AT-1004R	Open Path Gas Detector (Receiver)	IR Type Open Path HC Detector-Receiver
26	BBOU-MAC-1005A	Manual Alarm Call Point (Field ESD)	Manual Call Point
27	BBOU-AE/AT-1003	Point Type Gas Detector	IR Type HC Detector
28	BBOU-AE/AT-1004T	Open Path Gas Detector (Transmitter)	IR Type Open Path HC Detector-Transmitter
29	BBOU-MAC-1005A	Manual Alarm Call Point (Field ESD)	Manual Call Point
30	ELEP-HB-1005	Hooter and beacon combine	Hooter-Beacon